



Final Internship Report

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AI / ML Intern

Acknowledgements:

I would like to express my sincere gratitude to Embed Nexus for providing me with the opportunity to complete this internship. It has been a rewarding journey where I had the chance to expand my knowledge, apply classroom concepts to real-world challenges, and enhance my professional skills. I am especially thankful to my supervisors **Mr Awais** and **Mr Syed Kamran Hussain Shan** for their continuous guidance, encouragement, and support throughout the program. Their valuable feedback and mentorship have been instrumental in my learning process. Additionally, I had the opportunity to collaborate with like-minded peers on various projects, which gave me valuable insight into working in a professional environment.

Introduction:

This internship was designed to provide comprehensive, hands-on exposure in the fields of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL). The primary objective was to bridge the gap between theoretical concepts learned during coursework and their practical applications by engaging in real-world projects and collaborative problem-solving. Throughout the internship, I was able to contribute to a wide range of tasks, starting from foundational ML concepts to advanced experimentation with complex model deployment in Natural Language Processing (NLP). This structured progression allowed me to build confidence in both theoretical understanding and technical implementation.

The experience not only strengthened my technical knowledge but also enhanced my problem-solving, research, and professional communication skills. In the early phases, we were thoroughly familiarized with different **data preprocessing and cleaning techniques**, including duplicate removal, null imputation, feature scaling, and normalization. These activities helped me understand how raw datasets are transformed into usable formats suitable for machine learning models. Following this, we were introduced to **core ML methodologies**, where we learned how to partition datasets into training, validation, and testing sets, and the importance of maintaining fairness and avoiding data leakage.

We then applied different ML models, experimented with **hyperparameter tuning**, and explored how careful parameter selection directly impacts model performance, generalization, and accuracy. As the internship progressed, we moved to more advanced **Deep Learning concepts**, exploring neural networks and their applications. This provided exposure to ANN training, optimization challenges, and understanding how deeper architectures learn representations from data.

In the later stages, we explored **language and NLP concepts** such as stemming, tokenization, and preprocessing text data to make it machine-readable. Finally, we gained practical experience with **deploying Large Language Models (LLMs) locally**, which involved setting up environments, downloading model files, configuring pipelines, and experimenting with generation parameters. This end-to-end journey, starting from data cleaning all the way to LLM deployment, gave me a holistic understanding of the AI/ML workflow and real-world application scenarios.

Week 1:

During the first week, I focused on building a strong foundation in Machine Learning concepts. I explored:

- **Supervised learning:** Understanding regression and classification tasks.
- **Unsupervised learning:** Learning about clustering and dimensionality reduction.
- **Classification tasks:** Differentiating between binary, multi-class, and multi-label classification.

I also studied the **basics of supervised learning** in greater detail, including the need for labeled datasets, the role of input-output mapping, and how the model learns patterns from training data to make predictions on unseen data. I explored the importance of training and validation splits, the concept of overfitting and underfitting, and why supervised learning is essential for applications such as spam detection, sentiment analysis, and price prediction.

This learning was presented in my **First Week Presentation**, where I explained ML basics, supervised learning requirements, use cases, and applications in real-world scenarios.

Week 2:

During the second week, I worked with a dataset obtained from the **Global Health Observatory (GHO)** under the World Health Organization (WHO). This dataset tracked health status and related economic factors for 193 countries. The economic data was supplemented with information from the United Nations. For this project, only the most critical health-related factors were selected, focusing on the years 2000–2015. The merged dataset consisted of 22 columns and 2938 rows, representing 20 predicting variables divided into four broad categories: Immunization-related factors, Mortality factors, Economic factors, and Social factors.

At first, a visual inspection of the dataset revealed **missing values**. Since the data originated from WHO, there were no evident errors, but missing entries were noticeable in population, Hepatitis B, and GDP, mostly for smaller or less-documented countries such as Vanuatu, Tonga, Togo, and Cabo Verde. Using **Python**, I analyzed the missing data distribution with visualization libraries and exploratory data analysis techniques. Given the difficulty of filling in values for such countries, it was decided to exclude them from the final model dataset.

Once the final dataset was prepared, I performed a series of **data preprocessing tasks**, including cleaning, imputation, and fixing outliers. For imputation, different statistical methods were used to replace null values, and normalization techniques were applied to bring features to a consistent scale. Outliers were handled through a combination of capping methods and transformations, ensuring the dataset was suitable for model training.

After cleaning, I applied **machine learning models** on the dataset to predict life expectancy using the health, economic, and social predictors. I also carried out **hyperparameter tuning** to improve the performance of these

models, testing different configurations to achieve optimal accuracy and reliability. This provided me with strong hands-on experience in end-to-end data preprocessing and model training on real-world health data.

Week 3:

In this phase, I worked with datasets having different sampling rates, specifically **20 GSps** and **250 Msps**. Each dataset consisted of multiple features capturing the statistical and signal characteristics of the data. The columns included:

- **Steady_State**
- **Energy_of_Signal**
- **Energy_of_Envelop**
- **Variance_of_Signal**
- **Entropy_of_Instantaneous_Phase**
- **std_of_Signal**
- **slope_Energy_Distribution**
- **Energy**
- **Variance_of_Energy**
- **Max_sum_energy**
- **3rd_order_poly_coef_3**
- **3rd_order_poly_coef_2**
- **3rd_order_poly_coef_1**
- **3rd_order_poly_coef_c**
- **Device**

Both the 20 GSps and 250 Msps datasets shared this same structure, making them suitable for comparative analysis and model training. The data was collected using **Bluetooth at different frequencies**, which introduced variability in the captured signal features. These features captured essential information such as statistical variation, frequency components, and polynomial coefficients of the signals, which were then used as inputs to Artificial Neural Networks for training and evaluation.

In the third week, my focus shifted to working with **Artificial Neural Networks**. I conducted experiments on two datasets: **Msps** and **Gsps**.

- For the **Mspds dataset** (trained for 350 epochs):
 - Training Accuracy: ~51.9%
 - Validation Accuracy: ~54.2%
 - Test Accuracy: ~48.7%
- For the **Gspds dataset** (trained for 300 epochs):
 - Training Accuracy: ~9.3%
 - Validation Accuracy: ~8.8%

Through these experiments, I learned how to evaluate models using metrics like **accuracy** and **loss** on training, validation, and test sets. This helped me understand overfitting, underfitting, and the importance of hyperparameter tuning.

Week 4:

In the final weeks, I focused extensively on **deploying pre-trained Large Language Models (LLMs) locally** using the Hugging Face platform. This phase was crucial because it provided me with real-world exposure to Natural Language Processing (NLP) workflows and the deployment of state-of-the-art models. My work included:

- Learning about the core components of Hugging Face models, such as `config.json`, `tokenizer.json`, `model.safetensors`, and special token mappings.
- Setting up the environment with the required Python libraries like **Transformers**, **Torch**, and **huggingface_hub**.
- Downloading and storing models locally to enable offline usage and reduce dependency on external servers.
- Implementing pipelines for **text generation, classification, and summarization** while experimenting with parameters such as `max_new_tokens`, `temperature`, `top_k`, and `do_sample` to control the creativity and quality of outputs.
- Exploring the **advantages of local deployment**, including faster response time, greater control over models, and reproducibility of results.

Additionally, I practiced **fine-tuning models** with small custom datasets to adapt the models for specific tasks. This helped me understand how pre-trained knowledge can be extended and tailored to new domains. I also worked on error handling, environment optimization, and system performance monitoring to ensure smooth model execution. By the end of this phase, I had gained confidence in setting up, running, and customizing Hugging Face models in local environments, a skill that is highly applicable in research and industry.

Key Learnings

- Strong understanding of **ML fundamentals** and supervised/unsupervised learning.
- Practical experience with **ANNs** and model evaluation.
- Deployment and usage of **Hugging Face pre-trained models**.
- Enhanced technical skills in **Python, TensorFlow** and model pipelines.
- Improved problem-solving, teamwork, and presentation abilities.

Challenges Faced

- Handling large datasets and training limitations due to computational resources.
- Configuring dependencies and managing environments for Hugging Face models.
- Understanding trade-offs between model complexity, accuracy, and resource needs.

Overcoming these challenges gave me the ability to adapt quickly, troubleshoot effectively, and think critically.

Conclusion

This internship has been an **enriching experience** that significantly contributed to my academic and professional development. I was able to:

- Apply theoretical knowledge to practical problems.
- Gain hands-on experience in **AI, ML, and NLP**.
- Understand the workflow of model training, evaluation, and deployment.

I am confident that the knowledge and skills gained during this internship will be valuable for my future career in **Data Science and Artificial Intelligence**. I look forward to leveraging this experience in academic research, professional projects, and further studies.

Final Reflection: The internship at Embed Nexus has not only enhanced my technical expertise but also improved my adaptability, problem-solving, and professional growth. It has been a stepping stone toward my goal of pursuing advanced work in AI and Machine Learning.